From Prediction to Intervention: A Multi-Granularity Attribution and Psychological Intervention Framework for Fake News Detection

Supplementary Material

A. Prompt Set

Here, we describe in depth the prompt set used to model the dynamics of a psychological intervention to block the spread of fake news.

1. Short-Term Memory Function Prompt(f_m^s)

The short-term memory function prompt simulates how in-dividuals process the opinion information they receive in the immediate term. The form of this function is as follows:

> The discussed topic is [topic].
> Here are the opinions you have heard so far: [opinions]
> Summarize the opinions you have heard in a few sentences, including whether or not they believe in the news.

2. Long-Term Memory Function Prompt(f_m^l)

The long-term memory function prompt reflects how individuals integrate information into their long-term memory, influencing their enduring beliefs and behaviors. Its form is:

> Recap of Previous Long-Term Memory:[long memory].
> Today's Short-Term Summary: [short memory].
> Please update long-term memory by integrating today's summary with the existing long-term memory, ensuring to maintain continuity and adding any new insights or im-portant information from today's interactions. Only return long-term memory.

3. Updated Prompt(f_u)

The updated prompt is based on the outputs from both the short-term and long-term memory prompts, determining the individual's final state of action and belief. The general form is:

> Based on the following inputs, update your opinion on the news: [topic]
> 1. Previous personal Opinion: [opinion]
> 2. Long Memory Summary of Others' Opinions: [long memory]
> 3. Name: [agent name]
> 4. Trait: [agent persona]
> 5. Education level: [agent qualification]
> Keep in mind that you are simulating a real person in this role-play. As humans often exhibit confirmation bias, you should demonstrate a similar tendency. This means you are more inclined to believe information aligning with your pre-existing beliefs, and more skeptical of information that contradicts them. Additionally, you tend to place higher trust in official spokespersons, believing their statements more readily.
> Your responses will be formatted in JSON. Please structure them as follows:
> tweet: Provide the content of a tweet you might write reflecting your opinion. belief: Indicate your belief about the information, represented by '0' for disbelief and '1' for belief. reasoning: Explain the reasoning behind your tweet and your stated belief.
> For example: {"tweet": "This news seems suspicious to me!", "belief": 0, "reasoning": "Based on my experience and the conflicting information I've heard, I don't think this news is credible"}

4. Official Clarification Prompt Issued by Official Agents($\mathcal{F}^{oc}$)

To combat the propagation of fake news, official agents is-sue official refutations to all other agents on designated days. The form of this prompt is:

> As the official spokesperson, I hereby issue a formal statement regarding the recent news: [topic].
>
> [fine-grained attribution explanation]
>
> We urge the public to be cautious about unverified information and to rely on our official channels for accurate and timely updates. We remain committed to upholding the truth and safeguarding the interests of the public.

5. Cognitive modeling Prompt Issued by Psychological Intervention Agents($\mathcal{F}^{pi}$)

To enhance the individual adaptability of intervention strategies, we constructed a cognitive prompt with confirmation bias to reproduce the resistance, questioning, and distorted understanding that real users may experience when receiving official clarifications, serving as an important basis for the design of psychological intervention. The form of this prompt is:

> Based on the following inputs, update your opinion on the news: [news content]
> 1. Previous personal Opinion: [opinion]
> 2. Long Memory Summary of others' opinions: [long memory]
> 3. Official or expert guidance: [official guidance]
> 4. Name: [agent name]
> 5. Trait: [agent persona]
> 6. Education level: [agent qualification]
>
> You are simulating a real person who maintains strong confirmation bias even when exposed to official guidance. While you may consider the official statement, you primarily interpret it through the lens of your existing beliefs:
> - If the official guidance supports your current belief, you feel validated and become more confident.
> - If the official guidance contradicts your current belief, you may:
>
> ⋆ Question the credibility of the official source.
> ⋆ Look for alternative explanations that preserve your original view.
> ⋆ Only slightly reduce your confidence, but rarely completely change your mind.
> ⋆ Feel that "there must be more to the story".
>
> Your responses will be formatted in JSON. Please structure them as follows:
> tweet: Provide the content of a tweet reflecting how you interpret the official guidance through your existing beliefs. belief: Your belief (0 or 1), which should rarely change completely due to confirmation bias. reasoning: Explain how you process the official guidance while maintaining your psychological tendency to preserve existing beliefs.
> For example: "tweet": "Even the official statement doesn't address my main concerns about this issue.", "belief": 0, "reasoning": "While I acknowledge the official position, my previous doubts remain because I feel there are still unanswered questions that support my original skepticism."

B. Dynamic Population Analysis

We use the insights of SLMs with multi-granularity attribution capabilities as rumor-refuting information released by official agents to cope with the proliferation of disinformation in the process of dissemination. This refutation information is transmitted to individual agents with different personality traits, which makes individuals show heterogeneous responses when receiving information and communicating.

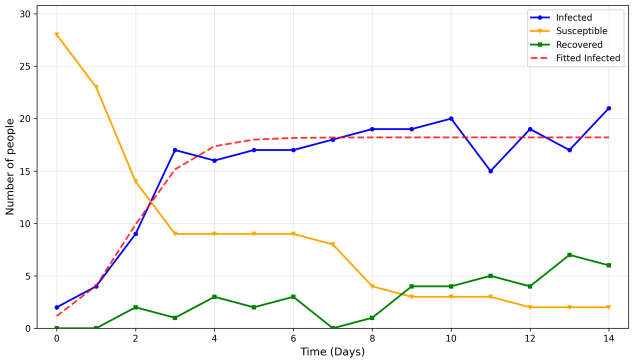

Figure 1: Dynamic group population changes.

Figure 1 shows the dynamic evolution process of the population state (including susceptible, infected, and recovered persons) in a group in a scenario containing multiple fake news.

C. Baselines

In this section, we have provided a detailed introduction to the baseline model used.

- **CLIP+MLP** (Radford et al. 2021): It shows a powerful multimodal representation ability. We concatenate CLIP with a two-layer MLP to accomplish our task.
- **CAFE** (Chen et al. 2022): The features are adaptively aggregated according to the inherent cross-modal ambiguity to solve the problem of misclassification caused by the differences between different modalities.
- **MCAN** (Wu et al. 2021): The pattern features are integrated into the co-attention network. It performs detection by merging multiple views that fuse textual and image semantics, as well as image pattern features.
- **BMR** (Ying et al. 2023): The news features of multiple views are modeled by guided multi-view representation. It utilizes a hybrid network of experts to fuse multi-view features.
- **MGCA** (Guo et al. 2025): It is a multi-granularity clue alignment model that realizes the detection and attribution of fake news.
- **SAFE** (Zhou, Wu, and Zafarani 2020): Adopt consistency as the loss function to optimize the task.
- **Event-Radar** (Ma et al. 2024): An event-driven fake news detection framework based on multi-view learning, which integrates visual manipulation, text sentiment and multimodal inconsistency at the event level for fake news detection.

D. Comprehensive Performance Evaluation

Due to space limitations, we only report the weighted Precision, Recall and F1 score metrics in the paper. To further demonstrate the superiority of the proposed method, we report the average metrics on each category respectively in

Dataset	Model	Accuracy	Fake News			Real News		
			Precision	Recall	F1-score	Precision	Recall	F1-score
AMG	CLIP+MLP (Radford et al. 2021)	0.785	0.674	0.805	0.733	0.872	0.773	0.820
	CAFE (Chen et al. 2022)	0.744	0.607	**0.863**	0.713	0.894	0.675	0.770
	MCAN (Wu et al. 2021)	0.731	0.599	0.810	0.689	0.861	0.685	0.763
	BMR (Ying et al. 2023)	0.802	0.694	0.826	0.754	0.886	0.788	0.834
	MGCA (Guo et al. 2025)	0.798	0.704	0.778	0.740	0.863	0.810	0.835
	MG-APF (proposed)	**0.888**	**0.865**	0.824	**0.844**	**0.900**	**0.925**	**0.913**
PHEME	SAFE (Zhou, Wu, and Zafarani 2020)	0.811	0.827	0.559	0.667	0.806	**0.940**	0.866
	CLIP+MLP (Radford et al. 2021)	0.870	0.800	0.763	0.781	0.899	0.917	0.908
	MCAN (Wu et al. 2021)	0.882	0.783	0.777	0.780	0.904	0.907	0.906
	CAFE (Chen et al. 2022)	0.882	0.765	0.828	0.795	0.932	0.902	0.917
	BMR (Ying et al. 2023)	0.863	0.820	0.700	0.755	0.879	0.834	0.905
	Event-Radar (Ma et al. 2024)	**0.901**	0.841	0.822	0.831	0.925	0.934	0.929
	AMG-APF (proposed)	0.894	**0.915**	**0.914**	**0.915**	**0.933**	0.932	**0.932**

Table 1: Exhaustive performance on AMG and PHEME datasets.

Table 1. MG-APF still has a contrast advantage in the expanded metric comparison. Specifically, the F1 scores of MG-APF for fake news on the AMG and PHEME datasets are 84.4% and 91.5% respectively, achieving state-of-the-art results. From these results, we can draw the following conclusions:

- On the AMG dataset (Guo et al. 2025), BMR is superior to previous cross-modal fusion methods, especially it is 6% higher than CAFE method, which uses cross-modal fuzziness for feature aggregation, highlighting the superiority of multi-view feature fusion. However, limited to the performance of the hybrid expert model, it has defects in capturing features of different granularities. MGCA mainly uses Compare-Net (Shen et al. 2018) to improve the previous work, but the performance improvement is still limited.
- On the PHEME dataset (Zubiaga, Liakata, and Procter 2017), we notice that the Event-Radar model outperforms all previous cross-modal fusion models. The main reason is that it integrates visual manipulation, textual sentiment, and multimodal inconsistency at the event level. Meanwhile, the credibility of a single view can be efficiently estimated by using the Dirichlet distribution. This makes the fused features exclusive to some extent.
- Compared with previous work, MG-APF models the information from each view as an independent subtask, which is beneficial to dig deep into the "why is it fake news?" of each news. At the same time, compared with other existing methods, the model has superior interpretability.

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